Arnav Mahajan

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Summary

Third-year Computer Engineering student at KJ Somaiya (Expected 2028) focused on backend, infrastructure, and systems engineering. Production experience with Go, PostgreSQL, AWS, CI/CD, concurrency, and distributed systems.

Education

KJ Somaiya College Of Engineering — Vidyavihar, Mumbai
Bachelor of Technology in Computer Engineering — *2028 Expected (8.68 CGPA)*

Experience

Open Source Contributor — October 2026 – Present
Meshery (CNCF) — Go, Kubernetes, GraphQl, PostgreSQL, NATS — *Remote, Flexible*

- Contributed to Meshery (CNCF) by resolving a 'mesheryctl' context-creation issue involving missing component-reference validation, improving CLI correctness; Actively contributing to its Go/Kubernetes cloud-native codebase through issue investigation, debugging, and platform improvements.

Backend Engineering Intern — February 2026 – July 2026
QueueFree — Golang, PostgreSQL, AWS — *Remote, Flexible*

- Architected and scaled a multi-tenant, real-time queue management platform in Go, coordinating order and operational workflows for 10,000+ active users.
- Engineered concurrent, event-driven backend services for queue scheduling, order lifecycle management, and real-time state propagation under concurrent workloads.
- Designed and deployed production AWS infrastructure across EC2, RDS, S3, and ECR, building CI/CD pipelines and operational workflows for reliable deployments.
- Migrated PostgreSQL workloads to AWS RDS and introduced table partitioning, reducing query latency by 35% and sustaining 2,000+ queries/sec under benchmark workloads.

Backend Developer Intern — July 2025 – Aug 2025
PearlThoughts — NestJs, TypeORM, PostgreSQL — *Remote, Fulltime*

- Engineered a modular NestJS medical scheduling backend with JWT/OAuth authentication and RBAC, building timezone-safe appointment and availability services with concurrent booking/conflict validation, time-slot constraints, and soft-delete workflows to ensure secure and consistent scheduling.

Projects

Veyra - ULPF | *Python, Go, Kafka, ClickHouse, Docker, Wazuh* — 2026

- Engineered an air-gapped, vendor-agnostic log processing platform that ingests heterogeneous Syslog/HTTP telemetry through edge collectors and Kafka, deterministically normalizes events to OCSF while preserving raw evidence and never silently dropping malformed logs.
- Designed a 10-service event-driven pipeline spanning ingestion, normalization, routing, archival, lineage indexing, integrity verification, and SIEM delivery, using Kafka for decoupled processing and ClickHouse for high-volume lineage/evidence queries.
- Built a tamper-evident evidence layer using hash-chained sealed segments, Merkle trees, and Ed25519 signatures, enabling byte-level traceability and verification of log integrity across forensic workflows.
- Benchmarked 1B+ events at 300K+ events/sec on a local workstation, with automated load testing, smoke checks, integration checkpoints, and hardware-aware deployment profiles for reproducible performance evaluation.

Re:Learn | *Python, LightGBM, Bayesian Inference, Adaptive Learning* — 2026

- Engineered a multimodal learning backend that analyzes learner code and responses to diagnose underlying programming misconceptions, combining model inference, rule-based analysis, and learner telemetry into targeted remediation workflows.
- Built a knowledge-graph-driven learner state system that tracks competency, misconceptions, and remediation progress, enabling adaptive reassessment and personalized learning paths across interactive coding missions.

Technical Skills

Languages: Golang, Python, C/C++, TypeScript, SQL, Python
Systems: Concurrency, Operating Systems, Distributed Systems, Networking, DBMS
Cloud & Infrastructure: AWS (EC2, RDS, S3, ECR), Docker, GitHub Actions, Kubernetes

Achievements

Solved 1,600+ algorithmic problems across CodeChef and LeetCode. Codeforces Pupil. Winner at multiple Competitive programming Contests. 2x Hackathon Winner.